Taylor's Theorem in $\mathbb{R}^n$

R. H. Turani

August 2026

Let $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_n)$ be the set of all variables represented by the multi-index α, such that $\alpha \in \mathbb{N}_0^n$, and let $H = (h_1, h_2, \ldots, h_n)$ be the set representing the variation of indices, where $|P - P_0| < \delta$ is a neighborhood of P_0, contained in the open domain $D \subset \mathbb{R}^n$, and let $P(\alpha) = (\alpha + H) \therefore P(a) = (\alpha_1 + h_1, \alpha_2 + h_2, \ldots, \alpha_n + h_n)$, where $P(a) \in D$.

Let the radius of the neighborhood be $\delta = \sqrt{|H|^2}$, where $|H|^2 = (h_1^2 + h_2^2 + \ldots + h_n^2)$. Consider the function $f : D \subset \mathbb{R}^n \to \mathbb{R}$, $F(t) = f(x_{1,0} + h_1 t, x_{2,0} + h_2 t, \ldots, x_{n,0} + h_n t)$, where t belongs to the open interval $I_\epsilon = t : -\epsilon < t < 1 + \epsilon \supset [0, 1]$.

When t varies in the interval $[0, 1]$, the point $P_t = P_0 + t(P - P_0)$ varies along a segment AB, excluding the endpoints A and B, and containing the segment PP_0. Under these conditions, it is clear that the Maclaurin expansion of order n for the function $F(t)$ holds:

$$f(P) = \sum_{|\alpha| \leq n} \frac{D^\alpha f(P_0)}{\alpha!} (P - P_0)^\alpha + R_n(P) \tag{1}$$

where $R_n(P)$ is the remainder of the expansion, with $R \to 0$ as $P \to P_0$, that is, $\lim_{P \to P_0} \frac{R_n(P)}{|P - P_0|^n} = 0$.